

Community Detection on Signed Graphs: A Bayesian MCMC Coupling Approach



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Outline

01

Motivation & the Sankararaman–Baccelli Model





Haplotype Assembly: Graphical Example

1. Biological Setup: Two Homologous Chromosomes (Haplotypes)

Humans are diploid: they have two copies of each chromosome (homologous chromosomes).

----- Haplotype 1
(Truth $\Sigma = +$)

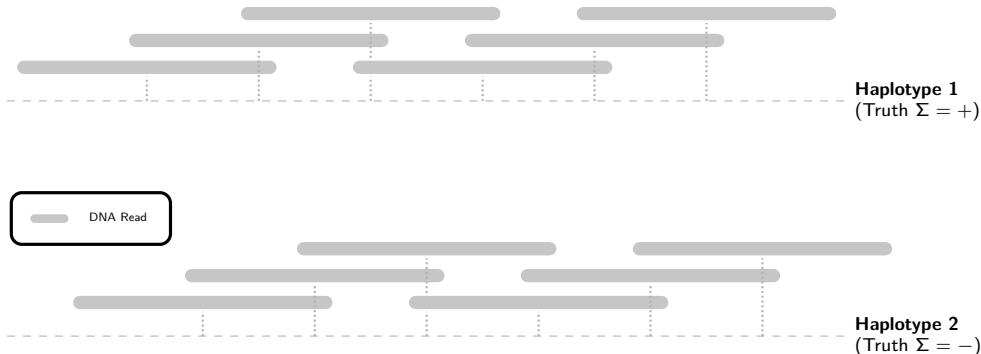
----- Haplotype 2
(Truth $\Sigma = -$)



Haplotype Assembly: Graphical Example

2. High-Throughput Sequencing: DNA Reads

Sequencing generates short, overlapping fragments called **reads** from both chromosomes.

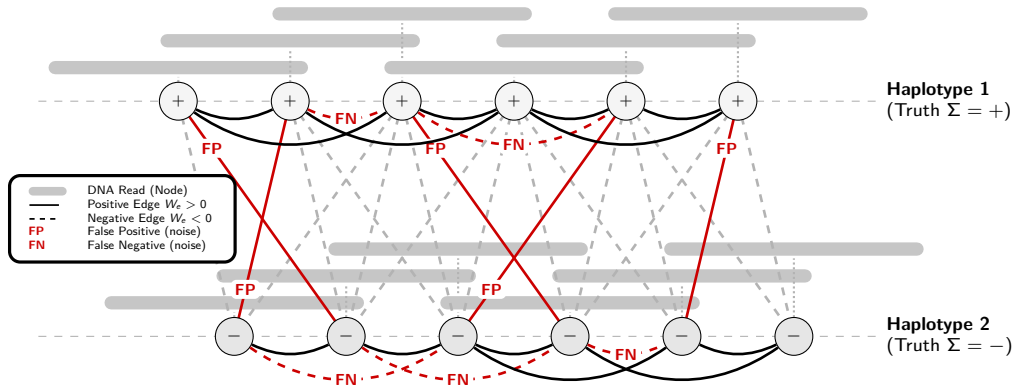




Haplotype Assembly: Graphical Example

3. Modeling as a Signed Graph

Nodes represent reads, and edges represent overlaps. Edge signs indicate match (+) or mismatch (-).





Motivation: Haplotype Assembly

Reconstructing Chromosomes from DNA Reads

- ▶ Humans are diploid: they have two copies of each chromosome (homologous chromosomes).
- ▶ **Haplotype assembly**: Reconstructing the sequence of nucleotides for each of the two chromosomes.
- ▶ High-Throughput Sequencing generates short, overlapping fragments called **reads**.

Mapping to Signed Community Detection

- ▶ **Nodes**: DNA reads, naturally mapped to genomic positions on $\mathbb{Z}$ (spatial structure).
- ▶ **Edges**: Overlap between reads (only overlapping reads share an edge).
- ▶ **Signs**:
 - **Positive edge (+)**: Reads match at overlapping polymorphic sites (same chromosome).
 - **Negative edge (-)**: Reads mismatch at polymorphic sites (different chromosomes).
- ▶ **Hidden Communities**: $K = 2$ partitions (Chromosome 1 vs Chromosome 2).



The Noisy Channel & Bayesian Formulation

From Sequencing Errors to Graph Weights

- ▶ Sequencing errors induce **frustration**: contradictions in pairwise labels.
- ▶ A sequencer has a known error rate ε .
- ▶ If two reads overlap at distance r_e , let:
 - $f_{\text{in}}(r_e) = \mathbb{P}(\text{overlap observed} \mid \text{same chromosome}) = 1 - \varepsilon$
 - $f_{\text{out}}(r_e) = \mathbb{P}(\text{overlap observed} \mid \text{different chromosomes}) = \varepsilon$

Natural Mapping to Signed Weights

- ▶ The log-likelihood of observation H under labels σ translates into signed weights:

$$W_e(H) = \begin{cases} \log \left(\frac{f_{\text{in}}(r_e)}{f_{\text{out}}(r_e)} \right) > 0 & \text{if } e \in H \text{ (observed overlap),} \\ -\log \left(\frac{1 - f_{\text{out}}(r_e)}{1 - f_{\text{in}}(r_e)} \right) < 0 & \text{if } e \notin H \text{ (absent overlap).} \end{cases}$$

- ▶ The posterior distribution is exactly a **Signed Gibbs measure**.

The Geometric Stochastic Block Model (GSBM)

Sankararaman & Baccelli (SODA 2018) [1]

- ▶ Nodes V_n have spatial positions in $\mathbb{R}^d$ (e.g., POISSON process).
- ▶ Latent community assignments $\Sigma \in \{+1, -1\}^{V_n}$ chosen uniformly at random.
- ▶ Connecting functions $f_{\text{in}}, f_{\text{out}} : \mathbb{R}_+ \rightarrow [0, 1]$ with $f_{\text{in}}(r) \geq f_{\text{out}}(r)$.
- ▶ An edge $e = \{i, j\}$ with distance $r_e = \|x_i - x_j\|$ is present in H with probability:
 - $f_{\text{in}}(r_e)$ if $\Sigma_i = \Sigma_j$ (intra-community)
 - $f_{\text{out}}(r_e)$ if $\Sigma_i \neq \Sigma_j$ (inter-community)

Goal: Weak Recovery

- ▶ Recover Σ up to label permutation strictly better than random guessing.
- ▶ Formulated via the **Community Overlap**:

$$\text{ov}_n(\Sigma, \tau) = \max_{\pi \in \mathfrak{S}_2} \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\{\tau_i = \pi(\Sigma_i)\}}$$

- ▶ Weak recovery is possible if there exists an algorithm τ achieving $\mathbb{E}[\text{ov}_n(\Sigma, \tau)] \geq \frac{1}{2} + \eta$ for some $\eta > 0$ as $n \rightarrow \infty$.



The Percolation Obstruction

Necessary Condition for Weak Recovery [1]

- ▶ Intuitively, if the graph is composed of many tiny, isolated components, no global community structure can be reconstructed.
- ▶ Information can only flow globally if the graph **percolates** (possesses a giant/infinite component).

Theorem (Sankararaman & Baccelli [1])

Percolation of the graph with edge probability

$$p_e = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

*is a **necessary condition** for weak recovery. If this independent bond percolation does not yield a giant component, weak recovery is mathematically impossible.*

- ▶ Our goal: Retrieve and extend this result using a unified **Bayesian MCMC coupling** framework.

02

The Bayesian Framework & Coupling





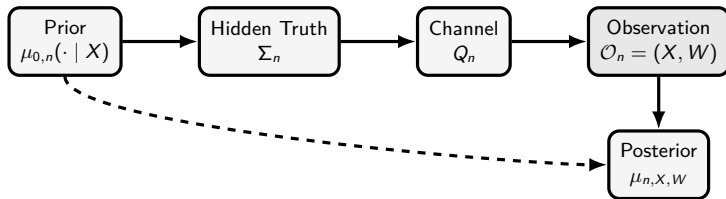
The General Bayesian Model

Definition

General Bayesian Community Detection Model

1. **Features:** $X_n \in \mathcal{X}_n$ with distribution ν_n .
2. **Prior:** $\mu_{0,n}(\mathrm{d}\sigma \mid x)$ over $\mathcal{C}_n = \mathcal{L}_K^{V_n}$.
3. **Channel:** $Q_n(\mathrm{d}w \mid x, \sigma) = \prod_{e \in E_n} Q_{e,n}(\mathrm{d}w_e \mid x_i, x_j, \sigma_i, \sigma_j)$.

The joint law is $\mathbb{P}_n(\mathrm{d}x, \mathrm{d}\sigma, \mathrm{d}w) = \nu_n(\mathrm{d}x) \mu_{0,n}(\mathrm{d}\sigma \mid x) Q_n(\mathrm{d}w \mid x, \sigma)$.





Posterior Resampling (Nishimori Identity)

Replacing Truth by a Replica

- ▶ Since Σ_n is hidden, we cannot compute the overlap $\text{ov}_n(\Sigma_n, \tau_n)$ directly in an algorithm.
- ▶ However, under the true model, the hidden truth and a sample from the posterior are exchangeable.

Theorem (Posterior Resampling / Nishimori Identity [2])

Let τ_n be any community detection algorithm. Let $\Sigma_n^{(1)} \sim \mu_{n, X_n, W_n}$ be a replica drawn independently from the posterior. For any bounded test function Ψ_n :

$$\mathbb{E} [\Psi_n(X_n, W_n, \Sigma_n, \tau_n)] = \mathbb{E} [\Psi_n(X_n, W_n, \Sigma_n^{(1)}, \tau_n)] .$$

Corollary

Overlap Identity For any algorithm τ_n and threshold $s \in [0, 1]$:

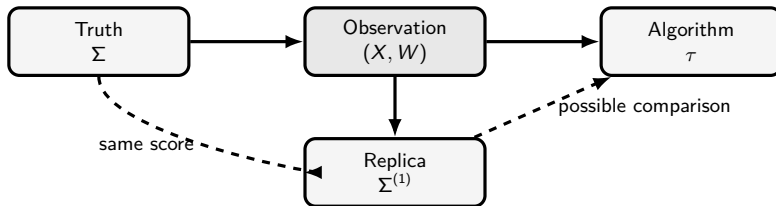
$$\mathbb{P} [\text{ov}_n(\Sigma_n, \tau_n) \geq s] = \mathbb{P} [\text{ov}_n(\Sigma_n^{(1)}, \tau_n) \geq s] .$$



Invariant Coupling

Markov Chain Dynamics as a Source of Coupling

- ▶ To study the overlap, we need a way to construct the posterior replica $\Sigma^{(1)}$ starting from the ground truth Σ .
- ▶ Let $M_{n,x,w}$ be a MARKOV transition kernel that leaves the posterior $\mu_{n,x,w}$ invariant.
- ▶ If we initialize the chain at the hidden truth Σ , and perform **exactly one step** of the dynamics to obtain $\Sigma^{(1)} \sim M_{n,X,W}(\Sigma, \cdot)$, then $\Sigma^{(1)}$ is still distributed according to the posterior!



03

MCMC Dynamics & Performance Bounds



MCMC Dynamics for Signed Gibbs Measures

1. Single-Spin Dynamics (Glauber / Metropolis–Hastings)

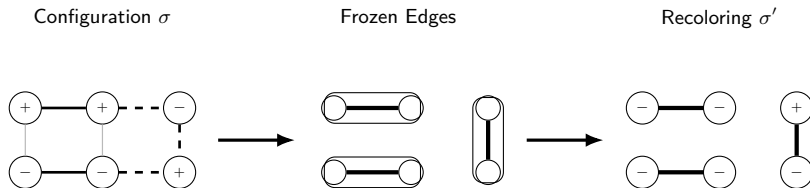
- ▶ Select a node at random and update its label according to its local neighbors.
- ▶ Extremely slow in the presence of frustration: local updates cannot cross energy barriers.

2. Swendsen–Wang (SW) Signed Cluster Dynamics [3, 4]

- ▶ Update whole clusters of vertices simultaneously to speed up mixing.
- ▶ **Freezing Step:** For each edge $e = \{i, j\}$:
 - If e is satisfied ($W_e > 0$ and $\sigma_i = \sigma_j$, or $W_e < 0$ and $\sigma_i \neq \sigma_j$), freeze it with probability:
$$p_e^{\text{gel}} = 1 - e^{-|W_e|}.$$
 - If e is unsatisfied, do not freeze it ($p_e^{\text{gel}} = 0$).
- ▶ **Recoloring Step:** Flip each frozen cluster independently with probability $1/2$.



One Step of Signed Swendsen–Wang



- Edges in bold represent satisfied interactions that are randomly frozen.
- Clusters are then flipped independently.



Coupling & the Sankararaman–Baccelli Bound

Gel Probability under the Generative Model

Let's run one step of SW starting from the hidden truth Σ . What is the probability that an edge e is frozen?

- **Case 1:** $\Sigma_i = \Sigma_j$. Edge is present with probability $f_{\text{in}}(r_e)$, and has weight $W_e = \log(f_{\text{in}}/f_{\text{out}}) > 0$.

$$\mathbb{P}(\text{frozen}) = f_{\text{in}}(r_e) \left(1 - e^{-|W_e|}\right) = f_{\text{in}}(r_e) \left(1 - \frac{f_{\text{out}}(r_e)}{f_{\text{in}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e).$$

- **Case 2:** $\Sigma_i \neq \Sigma_j$. Edge is absent with probability $1 - f_{\text{out}}(r_e)$, and has weight $W_e = -\log((1 - f_{\text{out}})/(1 - f_{\text{in}})) < 0$.

$$\mathbb{P}(\text{frozen}) = (1 - f_{\text{out}}(r_e)) \left(1 - \frac{1 - f_{\text{in}}(r_e)}{1 - f_{\text{out}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e).$$

Corollary

*Independent Bond Percolation Under the true generative model, the graph of frozen edges is **exactly** an independent bond percolation with edge probability:*



High-Order Triangular Dynamics

The Frustration Obstruction

- ▶ Standard edge-by-edge SW freezes satisfied edges independently.
- ▶ In frustrated systems, this leads to conflicts and blocks dynamics due to excessive freezing.
- ▶ **Idea:** Introduce correlations on the freezing mechanism using higher-order cliques (**triangles**) to freeze as little as possible.

3. Triangular Cluster Dynamics

- ▶ Transfer edge energy to triangles: $U(\sigma) = \sum_{t \in \mathcal{T}} U_t(\sigma)$.
- ▶ **Attractive Triangle** (+ + + or - - -): Freeze all 3 edges (prob $a_u = 1 - e^{-2u}$) or none (prob $b_u = e^{-2u}$).
- ▶ **Repulsive Triangle** (2 satisfied edges, 1 unsatisfied): Freeze exactly one of the two satisfied edges with probability $a_u/2$. This keeps the system in a low energy state with minimal freezing.



Triangular Dynamics Freezing Rules

$\triangle$ Attractive Triangle (Isotropic)					
Configuration	$\emptyset$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+, +, +)$	b_u	0	0	0	a_u
Other	1	0	0	0	0

$\triangleleft \triangle$ Repulsive Triangle (Isotropic)					
Configuration	$\emptyset$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+, +, +)$	1	0	0	0	0
$(+, +, -)$	b_u	0	$a_u/2$	$a_u/2$	0

Table: Freezing rules for triangular dynamics ($a_u = 1 - e^{-2u}$, $b_u = e^{-2u}$). By construction, these rules satisfy local reversibility and leave the posterior distribution invariant.



Strict Improvement on the Triangular Grid

Homogeneous Model ($f_{\text{in}} = p$, $f_{\text{out}} = 1 - p$)

On the triangular grid, we select a family of independent white triangles (like a chessboard).

- ▶ **Edge Dynamics (Standard SW):** Gelled graph is independent edge percolation with parameter $p_{\text{edge}} = 2p - 1$. Critical threshold: $p_c^{\text{edge}} = \frac{1}{2} + \sin(\frac{\pi}{18}) \approx 0.673$.
- ▶ **Triangular Dynamics:** Gelled graph yields correlated triangle percolation. Critical threshold: $p_c^{\Delta} = \frac{7}{4} - \frac{\sqrt{17}}{4} \approx 0.7192$.

Corollary

Strictly Better Impossibility Bound [5] Since $p_c^{\text{edge}} < p_c^{\Delta}$, for any

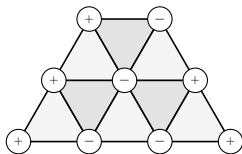
$$p \in [0.673; 0.7192[,$$

standard edge dynamics percole (no conclusion), whereas triangular dynamics do not. Hence, we prove that weak recovery is impossible in this wider range of parameters!



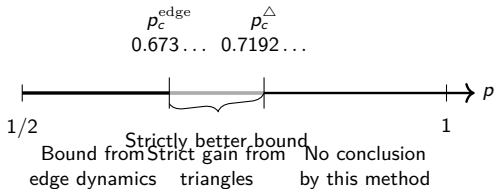
Comparison of Percolation Thresholds

Triangular Grid



White triangles used by
triangular dynamics

Weak Recovery Impossibility Bounds



04

Open Questions & Conclusion





Open Questions

1. Extension to Other Lattices and Higher Dimensions

- ▶ What are the thresholds on the Kagome, FCC, HCP, or Hyper-cubic lattices?
- ▶ Can we define tetrahedral or higher-order simplex dynamics to obtain even tighter bounds?

2. Multi-Community case ($K > 2$)

- ▶ How does the cluster dynamics extend when the spins take K values?
- ▶ What is the role of Potts-like percolation in the coupling?

3. Sufficiency of the Percolation Condition

- ▶ The percolation threshold provides a necessary condition (impossibility).
- ▶ Is percolation also sufficient for the existence of a reconstruction algorithm?



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Merci.

